CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in turning customers' requirements into working AI solutions, from agentic systems and RAG to AI evaluation and cloud deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by an AI consulting internship at Covestro and a strong foundation in machine learning and Python development.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer of an LLM-based knowledge-extraction system for the German energy sector, built end to end (data ingestion, chunking, prompt engineering, indexing with agentic, deterministic, and monolithic strategies); worked with SAP domain experts across the German energy sector and reported directly to the CTO.
- Built multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents in parallel) and evaluated it against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a web app (React frontend, Python microservices backend) on Azure where domain experts review and approve knowledge-graph changes, so AI output is checked by people before it is used.
- Built a Model Context Protocol (MCP) service exposing the knowledge graph with millisecond-latency retrieval, integrated into Claude Code for agentic development workflows.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, FastAPI, React, Neo4j, MongoDB, Postgres, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Advised business stakeholders on how agentic AI could support their decision-making workflows and translated their requirements into a solution approach.
- Designed, developed, and implemented the resulting agentic AI application with LangGraph, routing requests across vector database queries, general inference, and summarization.
- Built the full stack (React frontend, Python/FastAPI backend, API testing with Postman), containerized it with Docker, deployed it to AWS with CDK, and version-controlled it in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data.
- Built a Streamlit dashboard letting chemists simulate parameter changes and see predicted outcomes before running physical batches.

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- At a global IT services and consulting company, developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.

PROJECTS

Project: Meme Generation using LLMs | Open Llama 3B | QLoRA
- Fine-tuned an Open Llama 3B model with QLoRA to accelerate training, generating responses to educational meme-related prompts.

Project: Knowledge Graph | NLP (NLTK) | Protégé | SPARQL | RDF/OWL
- Developed a knowledge graph as a learning resource for AI students with Protégé and rdflib: information extraction with NLTK, RDF transformation, SPARQL querying, OWL enhancement.

TECHNICAL SKILLSET

- **Artificial Intelligence:** agentic systems (LangGraph, LangChain), LLM APIs (LiteLLM), RAG, GraphRAG, prompt engineering, MCP, AI evaluation (RAGAS, PytrecEval), fine-tuning (QLoRA), Claude Code
- **Software Development:** Python, SQL, FastAPI, REST APIs, React (beginner), Streamlit, Git, GitLab CI/CD, Docker, Kubernetes
- **Cloud & Data:** AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3), Azure, Postgres, MongoDB, Neo4j, vector databases
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)